

Bose Sullivan Kaikini

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OVERVIEW

Double major in Math/Computer Science specializing in Machine Learning and Pure Math, leveraging an interdisciplinary research and modeling background toward a career in Quantitative Finance

EDUCATION

2025-2027	University of South Carolina (Dean's Honor List)	Columbia, SC, US
2023-2025	Governor's School for Science and Math	Hartsville, SC, US
2020-2023	Dhirubhai Ambani International School	Mumbai, MH, India

EXPERIENCE & ENGAGEMENT

Internships

- **Research:** USC Engineering and Computing — Implemented topological data analysis techniques to analyze LiDAR plots through a National Science Foundation grant (Summer 2025, 20h/week)
- **Finance:** SC Federal Credit Union — Discount Cash Flow analysis and projections for overdraft policy overhaul impacting over 12k+ clients (Summer 2024, 40h/week)
- **Economics:** Newberry County Department of Economic Development — Customer acquisition and forecasting on cumulative project valued at \$10 million+ (Spring 2024, 10h/week)
- **Forecasting:** RSPN Equity Research — US mortgage and real estate market pricing and projections (Spring 2024, 10h/week)

Awards and Competitions

- **NASA SpaceApps Hackathon** — Winner - Best Use of Data
- **SC Quantum Quantathon** — Winner - Quantum Variational Machine Learning
- **Citadel University Applied Math Competition** — Top 3 - Modeling Unbiased State Voting Lines
- **SIMIODE Differential Modeling Competition** — Outstanding (perfect score) - Modeling synthetic-trained generative AI degradation
- **Rice University Datathon** — Track Winner - Ensemble ML prediction and signal analysis model
- **HackPrinceton** — Built urban issue classifier through glasses-mounted camera-based ML application
- **USC Magellan** — Grant Recipient - USC Premier Undergraduate STEM research grant
- **USC Journey** — Grant Recipient - Undergraduate STEM research grant

RESEARCH AND PROJECTS

- **Quantum Portfolio Optimization:** Developing continuous Quantum analogs of portfolio optimization techniques and determining ideal stock selection in Qiskit and Kuda as part of the USC AI Institute; integrating current classical MILP portfolio solver into a continuous quantum entangled optimization system, emphasizing variational learning algorithms and Wigner functions
- **Spectral Graph Theory:** Implementing computational approach to higher vertex values for DS graphs on Haemers' Conjecture, targeting a quasipolynomial bound for DS graph proportion; leveraging specific libraries and techniques including FLINT, Sage, Cloud/SLURM parallelism for efficiency while exploring Harary-Sachs, Walk matrix analysis and isomorphism invariants (paid)
- **Computational Rheology:** Modeling polymer elasticity and deriving constitutive equations from new dumbbell-based kinetic theory, focusing on the inverse Langevin function; analyzing the FENE, RS, Cohen closures through the conformation tensor over varying Deborah numbers while confirming via dumbbell-level (micro) simulations (paid)
- **Algotrading Personal Project:** Designing autonomous trading algorithms via Python and the Alpaca SDK utilizing Machine Learning and Stochastic Calculus, primarily focused on Futures markets; leveraged Scikit-learn and TensorFlow to build predictive models, employing Ornstein-Uhlenbeck processes for mean-reversion and Kalman Filters for hyperparameter estimation

TOP SKILLS

- Quantitative Modeling
- Data Analysis
- Machine Learning
- Dynamic Programming
- Python & C++
- Data Visualization

CAMPUS INVOLVEMENT

- USC Putnam & MCM Lead
- Student Gov. - Academics
- Hackathon Club President
- University Ambassador
- Algorithm Design TA
- Emerging Leaders Program

LANGUAGES

- English – Native
- Hindi – Native
- Spanish – Fluent
- Mandarin – YCT 4